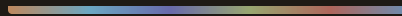


INTERNAL ASSESSMENT • 2026-08-23

What Lattice Actually Is

An evidence-backed audit — the inventions, the libraries, and the parts that do not hold up.



HOW THIS WAS BUILT

Eight agents read the repo before a slide was written.

Every claim carries a file path, a line number, or a number I ran myself. Nothing is quoted from the README without checking it against code.

220 findings, each with evidence

Every claim had to cite a path. Vague praise was rejected by contract.

92 of them are weaknesses

The brief said zero weaknesses means you did not look.

Zero were graded paradigm shift

All eight had the grade available. None used it.

What this covers, in order.

- 1 The shape of the thing — page 5
- 2 What is genuinely new — page 12
- 3 The paradigm question — page 25
- 4 Position against the field — page 29
- 5 Metrics, libraries, extraction — page 33
- 6 The autopsy, and where it goes — page 52

SECTION 01

The Shape of the Thing

VERIFIED • 2026-08-23

Lattice, by the numbers that are actually true.



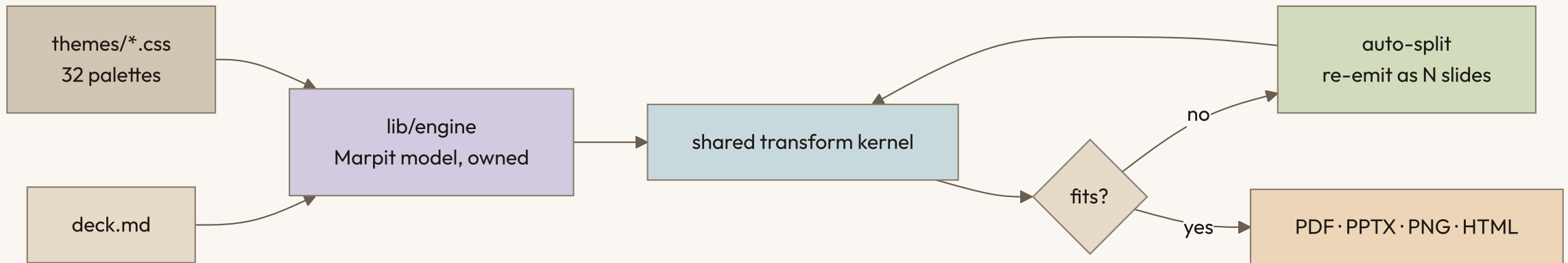
Where the code actually lives, measured this morning.

01	Engine and components	The render kernel, 61 manifests, the token system	88,617 LOC
02	Docs site and Studio	Astro, React, the Playground, five side libraries	130,126 LOC
03	Tests	313 unit files, 53 integration, 61 browser specs	83,750 LOC
04	Tooling	123 scripts, including one 9,998-line gate file	44,039 LOC
05	Decision archive	441 dated notes, growing about four a day	122,851 LINES

- Source lines only, excluding dependencies and generated output.

RENDER PATH

One owned engine, four formats, zero Marp.



THE BET

A slide is a compiled document, not a canvas.

The author declares semantics — a class name and plain Markdown. The engine owns every visual decision, including whether the content fits.

The mechanism is enumerable

61 manifests, 32 themes, one palette-blind stylesheet. Layouts never name a color.

The engine decides fit

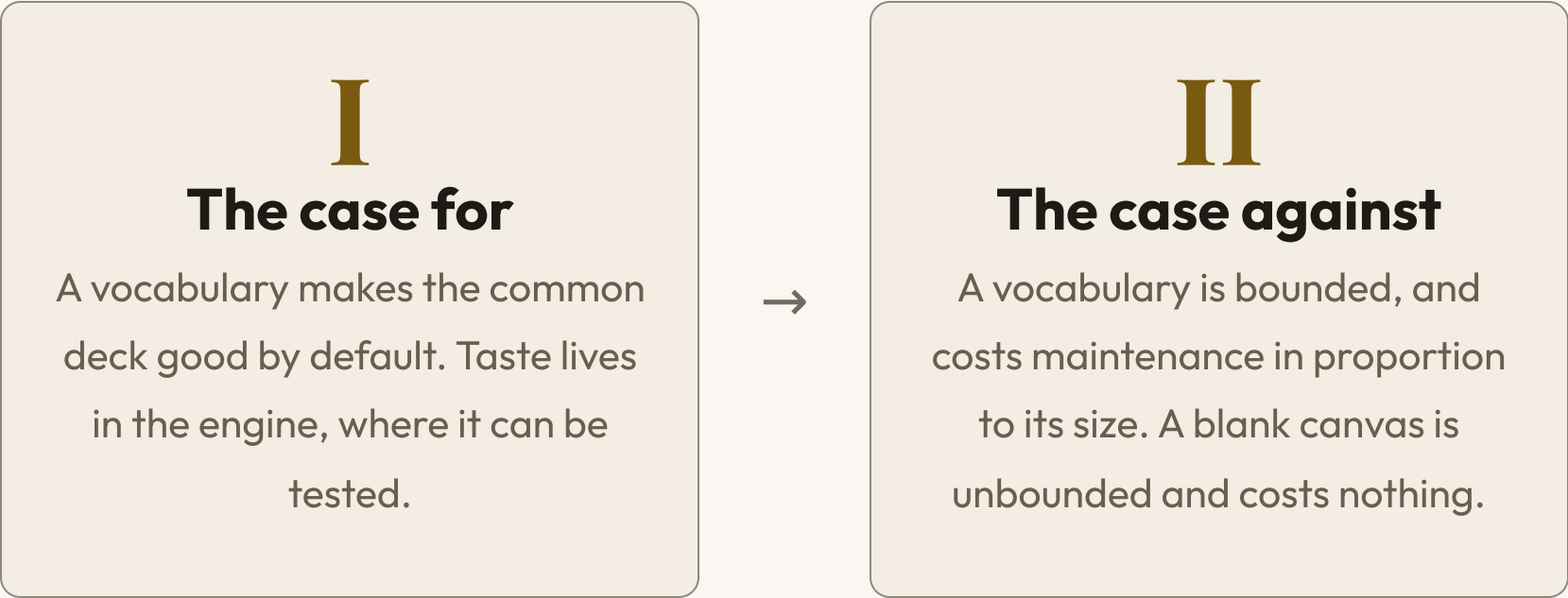
Slide count is derived from content and geometry. Authors do not paginate.

Verification is the product

Contrast and overflow are machine-checked against rendered pixels.

The bet has a real counter-argument, and it is not weak.

Both readings are defensible. The deck takes the second one seriously, because the room will.



♦ *Lattice did not remove the design work. It relocated it into 61 manifests that one person maintains.*

The escape hatch nobody documented.

The no-blank-canvas constraint is a strong default, not an enforcement. Front matter accepts a `style:` key that injects arbitrary CSS, and the Markdown parser runs with HTML enabled — so any author can bypass the component vocabulary entirely, on any slide, with no warning and no gate.

That is probably the right engineering call. It is not what the framing implies, and it appears in no authoring doc as an escape hatch.

SECTION 02

What Is Genuinely New

A Frame fits its content by collapsing, then shedding, then splitting — and if none of those make it fit at the readability floor, it shows an honest overflow. It never shrinks the type.

THE FIT SPINE · THE LOAD-BEARING INVARIANT OF THE ENGINE

SHRINK-TO-FIT PRIMITIVES

O

Every other deck tool answers “too much content” by making the type smaller. Lattice made that move structurally unavailable to itself.

The ban is on type, and it is code

No measure-then-shrink loop touches a font size. Chart geometry still letterbox-scales; glyphs never do.

The ladder is closed at four moves

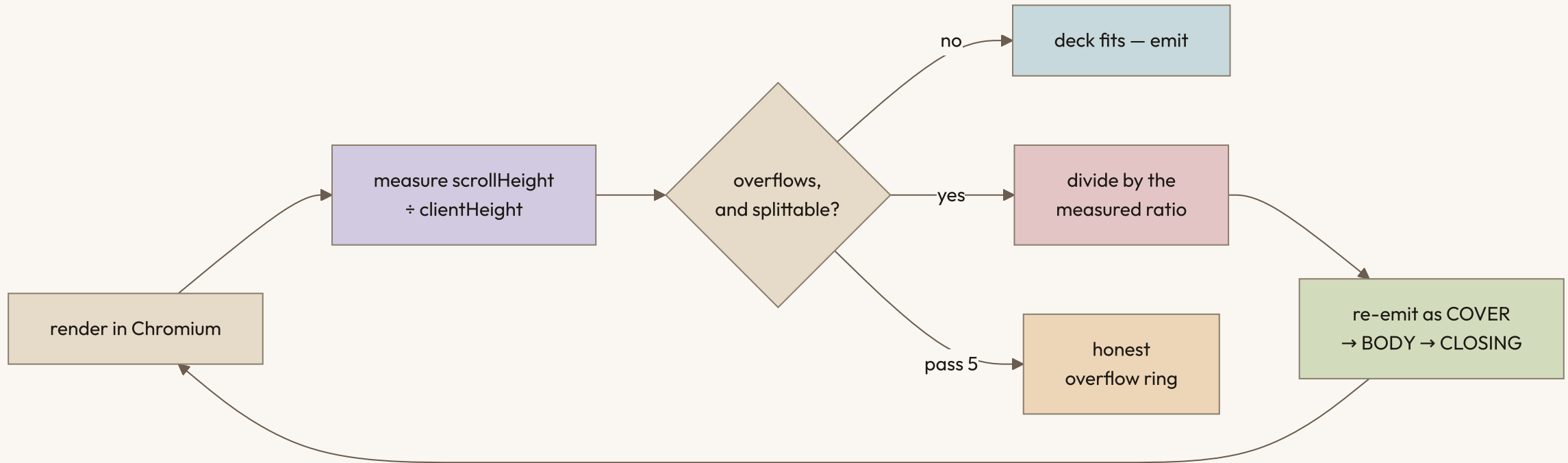
Collapse, shed, split, honest overflow. There is deliberately no fifth.

Readability is an axiom

A slide smaller than legible has failed, even if it fit.

AUTO-SPLIT

Pagination is measured in a browser, not counted in a parser.



Shrink-to-fit shipped here, then was purged on principle the next day.

June 21 to June 22, 2026

BEFORE

A token scaled prose to 0.66 in portrait. It worked, it was cheap, and it needed no measurement at all.



AFTER

The token was deleted. A headless render now measures real overflow and cuts the slide instead.

The mechanism was not broken. It was deleted for violating an axiom — a materially different, and much rarer, reason.

The large-text rule breaks here

The same type token resolves to 21.4px small and 64.1px large.

The obvious fix was built and deleted

Normalizing out the scale re-broke twenty committed portrait decks.

The answer was to exceed the standard

A flat 4.5:1 everywhere. It cost zero runs, measured.

CONTRAST RUNS FLIPPED BY SIZE

1,024

Rendering the same gallery at two canvas sizes flipped 1,024 of 1,534 contrast runs and 317 verdicts — on the size line alone, with no CSS change.

CONTRAST AUDITING

The audit reads pixels, not token pairs.

A token-pair audit passed 704 pairs green while the rendered deck carried 44 sub-AA runs, one at 2.54:1. So the gate moved to the DOM.

It finds the real background

Ancestors are walked through transparent paints to the painted surface.

It composites opacity groups

A blind spot that once reported 5:1 where pixels measured 3.21:1.

It reads generated text

Counters and badges are scored like any other run, every pull request.

COLOR

The ink solver fixes hue and moves only lightness.

lib/theme/cat-ink.js — one value clearing AA on both canvases

```
// A categorical mark has ONE hue across the deck. Its text-legible
// variant must clear WCAG AA against the light canvas AND the dark
// canvas — not "one value per canvas", which breaks hue identity.
//
// Hue and chroma are held EXACTLY; only OKLCH lightness moves.
const ink = solveInk(markHex, { light: lightBg, dark: darkBg, floor: AA });

// Leonardo, culori, chroma-js and Radix all solve against ONE
// background at a time. Solving both at once is the move that
// makes a palette-blind layout possible at all.
```

Ink means text

Scored at the 4.5:1 floor. No exceptions, no registration.

Mark means a shape

Scored at the 3.0:1 non-text floor.

Fill means a surface

Something to score against, never something to score.

TOKEN CONTRACTS

A token's contrast floor is derived from its name.

About a dozen name patterns replace a 383-entry lookup table — so a token added tomorrow classifies itself with no table edit.

CATALOG COST FOR AN AGENT

3.8k

The component catalog an AI agent must read before writing any slide fell from about 95,000 tokens to 3,800 — small enough to hold all 61 at once.

The compression was measured, and it cost

Top-1 retrieval fell 59.8% to 42.0% against the full prose, on 264 cases. Shipped anyway.

A separate ranker paid it back

A local search index moved top-1 from 16% to 86% on intent phrasings.

The fancier design was refuted

A weight tuner zeroed every signal in the facet scorer meant to replace it.

The security work produced one transferable idea and two unusual gates.

The serializer corollary

A closing style tag inside a CSS comment still ends the element — and any serializer round-trip silently undoes the escape.

A provenance census

Where a guard is impossible, each sink declares where its markup comes from, with an exact count.

The census found a hole

Two sinks certified as already-sanitized decoded entities on read. Both are now deleted.

KEY INSIGHT

Every arm of this rule was added after a real miss, not designed up front.

Four novelty claims, graded against hostile review.

Check holds · dash partial · cross fails

Component vocabulary

-  Unprecedented
-  Well executed
-  Category-defining

Beamer, Deckset, Sliderv, and Marp's own class mechanism — which is the one used here.

Two-surface ink solve

-  Unprecedented
-  Well executed
-  Category-defining

Every incumbent solves one background at a time. Narrow, real, defensible.

Rendered-DOM contrast

-  Unprecedented
-  Well executed
-  Category-defining

Composited auditing exists in research tooling. Gating a build on it does not.

Measured auto-split



-  Unprecedented
-  Well executed
-  Category-defining

Beamer's frame-breaks are honest prior art. No browser tool repaginates on fit.

READERS AWARDING "PARADIGM SHIFT"

A large, bold, dark blue letter 'O' centered on the page, serving as a decorative element for the text below.

Eight agents read 441 decision records and every benchmark. All eight had the grade available. None used it — and twenty-three findings came back graded genuinely novel instead.

SECTION 03

The Paradigm Question

Is "vocabulary instead of canvas" a paradigm shift?

The honest version of this argument has to survive someone who already knows LaTeX.

I

Yes, arguably

No prior tool makes fit a machine-checked property.

Content that does not fit becomes more slides, decided by measurement.



II

No, on the evidence

The vocabulary has four precedents, one of them Marp's own class mechanism — the exact mechanism Lattice uses.

✦ *Design systems have paired tokens, components and accessibility gates for a decade.*

RULING

Which half of the claim is load-bearing?

The vocabulary is the visible half and the derivative half. The verification is the invisible half, and the original one.

THE VOCABULARY FRAMING

- Reaches the widest audience
- Has four strong precedents
- Loses every argument with Beamer

THE VERIFICATION FRAMING

- Narrower, harder to explain
- Survives a hostile reviewer
- No competitor does it at all

RECOMMENDATION

Lead with verification. The vocabulary is table stakes; compiled fit and contrast are not.

SECTION 03 • RECAP

Where the shift is real, stated precisely.

- 01** Slide count is derived from content and geometry, not authored. → page 14
- 02** Contrast is a property of rendered pixels, gated per pull request. → page 17
- 03** Shrink-to-fit is structurally unavailable, so illegibility cannot be the fix. → page 13
- 04** The component catalog is a machine surface, benchmarked for retrieval. → page 20
- 05** Everything else is excellent execution of ideas that already existed. → page 25

SECTION 04

Position Against The Field

What each tool has that Lattice does not.

Left column audited · right column recalled

TOOL	LATTICE'S EDGE VERIFIED	WHAT IT HAS THAT LATTICE LACKS UNAUDITED
Marp	Components, tokens, measured split	User base, real editor preview
Slidev	Verification layer, agent catalog	Click builds, a hot-reload dev server
Beamer	Sub-second render, swappable palettes	Overlays, citations, cross-refs
Quarto	The design layer it leaves to a theme	Executable code chunks
PowerPoint	Text-diffable source, deterministic output	Co-editing, an editable artifact

◆ *This is the one slide in the deck that breaks its own rule: the right column is recalled, not checked against a running install. Treat it as the hypothesis to test, not the finding.*

Three things every competitor has that Lattice structurally does not.

A recipient who can edit

The PowerPoint export is one flat image per slide. It presents; it does not open.

Live collaboration

Collaboration is git. No co-editing, no comment thread, no presence.

Incremental build

No click-through reveals and zero transition rules in the engine. A speaker view does ship.

KEY INSIGHT

Deliberate scope choices — but a room will read them as gaps.

The independence claim is true, and narrower than it sounds.

Zero Marp packages ship, and the owned engine renders every first-party path. That part is verified and holds exactly as its own scorecard states.

Format independence is not achieved

The repo's own inventory finds 839 tracked files carrying a Marp reference, across 3,840 lines.

The compatibility tax is mispriced

The doc prices the runtime mirror at 2,182 lines. It is 3,251, and the doc says to re-measure.

The speed claim is frozen

Last measured June 2026 with Marp installed. It cannot be reproduced now.

SECTION 05

Performance, Measured Honestly

COMMITTED BASELINE

The engine numbers that are real.

2,519

Slides per second, stress set

GATED ON WORKLOAD



51.6 ms

58-slide deck, cold render

ABOUT 66% FIXED COST

113.9 S

6,967 unit tests, 0 failing

RE-RUN 2026-08-23

SIGNALS IN ONE EXIT CODE

2

The committed benchmark splits one exit code into two independent signals — the quietly cleverest thing in the measurement layer.

Workload fails everywhere

A moved slide count fails on any machine. That is a fact about code.

Timing gates almost nowhere

Only on a matching fingerprint with a calibration probe inside $\pm 15\%$.

It was born from real rot

A row sat blessed at the wrong count for a month while the check exited clean.

VERIFICATION

Run twice in one hour, it returns opposite verdicts.

node test/benchmark/engine-bench.mjs --check — both runs, verbatim

```
RUN 1 · probe 6.27ms vs blessed 4.92ms (27% off, band ±12%)
  same machine fingerprint — linux/x64, 4x Xeon @2.80GHz, node v22
  NOT COMPARABLE — timing is REPORTED, not gated.

RUN 2 · probe 4.59ms vs blessed 4.92ms (0.93x, inside band)
  same machine as the baseline — wall clock GATES
  dataset      base ms  now ms   Δ%   verdict
  normal (jargon)  51.61  64.69  +25.3 REGRESSION
  charts         43.52  57.70  +32.6 REGRESSION
  stress (jargon x6) 138.14 187.58 +35.8 REGRESSION
```

— Same code, same box, one hour apart. The refusal is doing real work; on a shared cloud runner it is also the only thing standing between this gate and three false regressions.

The print optimization is real, and its own numbers contradict its description.

58-slide deck · full versus re-place

AS DESCRIBED

Caching “the expensive half” and re-running “the cheap, DOM-free half”. The bench header prints “should be a fraction of full”.



AS MEASURED

The committed baseline: 121.7s full against 80.5s re-placed on the same 58 slides. Re-place is 66% of full — a 1.51× win, and the “cheap” half is 80.5s of it.

Worth keeping. The framing survived because nobody re-read the blessed numbers beside the sentence describing them.

OF A RENDER, THE FIXED COST

66%

Solve the two blessed rows as a fixed cost plus a per-slide cost and the headline "slides per second" is two thirds constant. It measures in-process render only; the rasterize pass that dominates a real export runs under a separate opt-in flag.

What is measured, and what nothing watches.

✓	Unit suite, build freshness, lint — every pull request	gated
✓	Rendered contrast and overflow invariants	gated
✓	Preview work budget — one keystroke, one render	gated
✓	Engine throughput and preview ceilings — head vs base, nightly	gated
—	Print, sweep, equivalence, quality baselines — blessed, unwatched	on-demand
○	Bundle size — no ratchet on any built artifact	ungated
○	359 committed PDF goldens, 148 MB, no cadence	ungated

— The nightly compares head against base on one runner. Four other blessed baselines are watched by nothing.

SECTION 06

The Libraries That Spun Out

Six are called libraries. Four are packaged. Zero published.

01	Vetrina	Self-driving walkthroughs; a cursor that never fakes clicks	PACKAGED
02	Cadenza	Caption and word-timing tracks, subtitle output	PACKAGED
03	Suono	Audio sequencing with pause-gating and mobile unlock	PACKAGED
04	Lente	Audience lenses — one source, several projections	PACKAGED
05	Anima	Chart and scene motion, vector-first	NOT A PACKAGE
06	Compose	The document model behind the Studio editor	NOT A LIBRARY

VETRINA

The one a stranger would plausibly install.

A self-playing product tour where a fake cursor drives the real application. The central promise is not marketing — it is verifiable in code.

It never fakes a click

Zero synthetic events across 2,890 lines. State moves through the host's own setters.

That design buys clean take-over

Because the engine emits no input, the first real event is unambiguously the user.

The prior art is a different thing

Shepherd, intro.js and driver.js highlight and wait for Next. None drives the app.

Standalone viability, graded without kindness.

Lente

- ✖ General problem
- ⊖ Decoupled
- ✖ Edge over prior art

The read path is 86 lines; the Studio glue is 541. Should not spin out.

Suono

- ⊖ General problem
- ✔ Decoupled
- ⊖ Edge over prior art

Best scheduler of the set. But it does not stream, which is why anyone would want it.

Cadenza

- ✔ General problem
- ✔ Decoupled
- ⊖ Edge over prior art

Already required by name from outside. Structurally Latin-script only, undisclosed.

Vetrina



- ✔ General problem
- ✔ Decoupled
- ✔ Edge over prior art

Real, unusual, well engineered. Blocked on packaging and license, not merit.

THE SHARED DEFECT

One stray key ships raw source to legacy resolvers.

docs/src/lib//package.json — verbatim, all four identical*

```

{
  ".": {
    "types": "./index.ts",           // ← the defect
    "import": { "types": "./dist/index.d.ts", // already correct
                "default": "./dist/index.mjs" },
    "require": { "types": "./dist/index.d.ts", // already correct
                 "default": "./dist/index.cjs" }
  }
}
```

```

// node16, nodenext and bundler read the NESTED conditions and
// resolve the shipped .d.ts. Only legacy node10 reads the top
// key — and gets raw .ts compiled under the consumer's tsconfig.
// The fix is deleting one line, not repointing it.
```

Three blockers stand between these libraries and npm.

Packaging, about a day

Fix the exports order, add a license file to each package, add repository fields.

Distribution, unproven

The release step publishes only the root package. The template has never been tested.

License, unresolved

All four are AGPL-only, which is a non-starter for most commercial adopters.

KEY INSIGHT

Every gate keeping these decoupled lives in the host repo and does not travel with the package.

PACKAGES PUBLISHED TO NPM

0

Version 1.0.0, zero git tags, and a release workflow that skips publishing when the token is unset. The documented install path does not work today.

SECTION 07

What Could Be Extracted Next

Three real candidates. The brief hoped for six to twelve.

The color kernel

2,240 lines, zero runtime dependencies, imported from 70 files. Already an internal library.

The crash sentinel

1,956 lines with zero imports of any kind. Coupled by six branded strings.

The input-verb kernel

493 lines — a 376-line DOM-free vocabulary plus a 117-line attachment layer.

KEY INSIGHT

Most monorepos contain zero extractable libraries. Three is a real result, not a thin one.

Two surfaces at once

It searches lightness until one value clears AA on both canvases.

Floors derived from names

A dozen name patterns replace a 383-entry table.

One coupling to sever

It hardcodes Lattice's token vocabulary. Needs a schema parameter, not a file move.

RUNTIME DEPENDENCIES

O

The strongest extraction candidate solves a problem the incumbents do not attempt, with no dependency surface at all.

The most-praised file in the repo is not a product.

WHAT IT LOOKS LIKE

A 9,998-line gate enforcing 70 checks — hex literals, margins, cascade layers, four injection-sink shapes — with allowlists that fail when stale.



WHAT IT ACTUALLY IS

No registry, no plugin interface, no ratchet helper. The one transferable idea is copy-pasted eleven times and runs to twenty lines.

Excellent infrastructure and a bad library. Worth keeping straight before anyone tries to package it.

SECTION 08

The Autopsy

Where the weight sits against what it defends.

Effort invested · value defended

High effort · High value

The verification layer, the gate file, the 6,967-test suite

Low effort · Low value

Fluent facades their own authors call “pure sugar”

Low effort · High value

The 3.8k-token pick surface; contrast floors derived from token names

High effort · Unclear value

441 decision notes, and five libraries serving the Studio rather than the engine

Five adoption blockers, and one by design.

- ☐ Install weight exceeds 1.2 GB, and the README understates it blocking
- ☐ Nothing has ever shipped — no tags, publish step skipped blocking
- ☐ Public docs are 13 pages against 441 internal records thin
- ☐ Bus factor is one, and the operating model assumes agents structural
- ☐ The repository is 236 MB, of which 148 MB is PDF goldens growing
- ☒ The learning curve is real — 61 components, each contracted by design

— The README says install “takes ~16s and needs no browser”. True of the build; not of the install.

INTERNAL PROSE PER PUBLIC PAGE

11k

There are 142,965 lines of internal engineering prose behind 13 published pages. The knowledge needed to use Lattice well lives in files the package does not ship.

The reference does ship

All 70 component contracts, 11,784 lines, plus the pick surface.

The operating manual does not

27 hard rules and 441 records stay in the repo, where no consumer sees them.

This is the most fixable blocker

Nothing about it is architectural. It is a writing backlog.

The canonical docs have drifted from their own artifacts.

01	Design system	Says 53 components in three places; the catalog says 61	STALE
02	Architecture	Calls the renderer 1,000 lines with no build system; it is 4,569 with three	WRONG
03	Competitive analysis	Marked shipped, cited as source of truth, states the wrong license	WRONG
04	Independence scorecard	Its own tax figure has drifted twice; it tells readers to re-measure	DRIFTING
05	Line-endings gate	Documented as shipped in five files; appears zero times in the gate	ABSENT

BY THE REPO'S OWN ACCOUNTING

Overbuilt, measured in lines that defend nothing.

15,838 lines.

Five side libraries serving the docs Studio, not the deck engine.

3,004 lines.

The Form catalog, from which the render path reads two fields.

1,428 lines.

A perf instrument: three retracted findings, one real leak — fixed in twenty lines.

217 lines.

A texture module with a “measured, not invented” preamble and no consumers.

Four claims the repo cannot currently back.

- ☐ “Every format is pixel-identical” — vector and raster are different artifact classes README
- ☐ “Passes WCAG AA throughout” — the gate exempts a documented decorative tier README
- ☐ “53 of 53 invariant coverage” — actually 54 of 61, opt-in, with no completeness gate stale
- ☐ “3–5× faster than Marp” — frozen in June, unreproducible, and the corpus has changed frozen

— And two it discloses itself: real-device behavior on phones is marked UNVERIFIED throughout, and the audio library says plainly that its browser path is tested against a mock. Hard rule 23 demands an artifact from the real surface, and the repo marks its own gaps more often than it papers over them.

The gates have gaps the gates cannot see.

Nineteen never fire

19 of 70 checks are named nowhere in the test suite, while the test claims full coverage.

One misses the likely case

The stylesheet gate does not walk the commonest document-assembling file type it scans.

A ratchet sits at 1,293

A rule stated as “target zero” tolerates 1,293 known violations.

KEY INSIGHT

These are text matchers whose own authors document that they cannot do dataflow.

What a hostile reviewer would ask first.

01 If it is this good, why has nothing shipped?

Publishing was never the goal; the engine was. It is a token and a workflow edit.

02 Can anyone but you maintain this?

Not today. The manual is 27 hard rules and assumes agent-driven throughput.

03 Is the decision archive an asset or a liability?

Both. It is the knowledge substrate, and its teardown process has never run once.

04 What breaks first at scale?

The docs. Everything internal is gated; everything external is 13 pages.

SECTION 09

Where It Goes

DIRECTION OF TRAVEL

The code says the next product is narrated presentation.



What would actually move the project.

PHASE 01

Now

weeks

DISTRIBUTION

☒ Fix exports order

LICENSING

☒ Decide per package

DOCS

☐ Correct the counts

VERIFICATION

☒ Contrast gated

PHASE 02

Next

a quarter

DISTRIBUTION

☐ Publish the engine

LICENSING

☐ Re-license

DOCS

☐ Write the guides

VERIFICATION

☐ Unwatched baselines

PHASE 03

Later

beyond

DISTRIBUTION

☐ Publish libraries

LICENSING

☐ Contributor terms

DOCS

☐ Reference parity

VERIFICATION

☐ Real-device tier

Three forks are open, and only one is technical.

Licensing gates distribution; distribution gates everything else

LICENSE

AGPL is right for the engine and wrong for four libraries whose entire value is being installed by strangers.

Leave it and the extraction work in section 07 is a hobby. Change it and you cannot change it back.

AUDIENCE

The engine, the Studio, and the library set are three products with three different buyers.

The repo currently optimizes for all three, which is why the docs serve none of them well.

TRANSFERABILITY

Either the operating model gets written down for humans, or it stays a single-operator system. Both are legitimate. Only one of them survives you being unavailable for a month.

The next ninety days, in unblocking order.

STEP 01

Answer the license question

Including answering "no change". It gates everything below it, and it is the slowest to reverse.



STEP 02

Make publishing possible

Delete the stray exports key, add per-package licenses and repository fields. No publish yet.



STEP 03

Correct the record

Four drifted docs, an absent gate cited in five files, one frozen benchmark claim.



STEP 04

Then publish, once, deliberately

Tagging is reversible. An npm name and a support expectation are not.

THE HONEST SUMMARY

What is the one-sentence version?

Both readings are true. The deck fails if it delivers only one of them.

THE GENEROUS READING

- An unusually rigorous engine
- Real inventions in fit and contrast
- Verification above category norms

THE ACCURATE READING

- The inventions are narrow and real
- The framing around them overstates
- It has never shipped to anyone

BOTTOM LINE

The engineering is better than the pitch, and the pitch is what is failing.

WHAT THIS IS FOR

The engine is done arguing with itself.

*Nothing here is a reason to slow down — it is the four cheap
things between a good engine and someone else using it.*

